A BOUNDED-EXPONENT ABELIAN GROUP WITH MAXIMAL D-TORSION IS CYCLIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The purely group-theoretic engine behind the cyclicity statement for mu_n(F) that opens the paragraph above the lemma.

Editorial note. The label lemma-cyclic names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Structure theorem for finite abelian groups: a finite abelian group decomposes as Z/e_1Z + ... + Z/e_rZ with 1 < e_1 | e_2 | ... | e_r
- Divisor sum identity: the sum of d over the divisors d of n equals n; for e_1 dividing each e_i, the e_1-torsion of the factor Z/e_iZ has exactly e_1 elements, so the e_1-torsion of the direct sum has e_1^r elements

Lemma 0.1. *Let A be an abelian group of exponent dividing n such that $\{x \in A \mid dx = 0\}$ has cardinality at most d for all $d|n$. Then A is cyclic of order dividing n.*

Proof. The conditions imply that $|A| \leq n$, in particular A is finite. The structure of finite abelian groups shows that $A = \mathbf{Z}/e_1\mathbf{Z} \oplus \ldots \oplus \mathbf{Z}/e_r\mathbf{Z}$ for some integers $1 < e_1|e_2|\ldots|e_r$. This would imply that $\{x \in A \mid e_1 x = 0\}$ has cardinality e_1^r. Hence $r = 1$. □

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